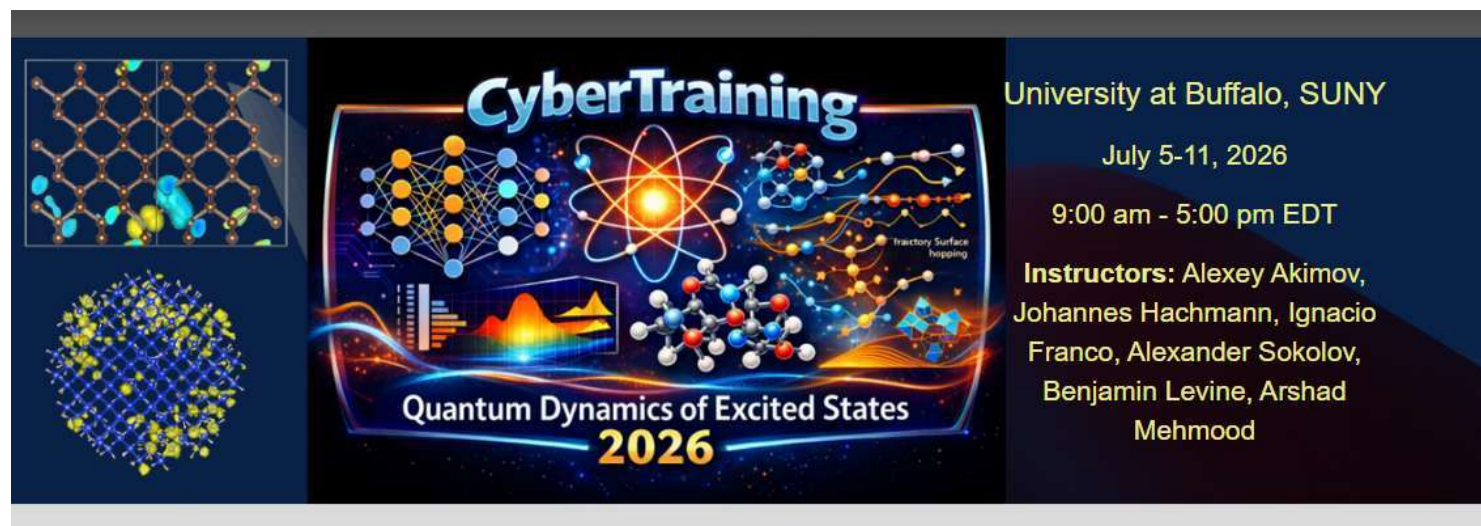


## Capstone Project



The NAMD study of all inorganic perovskite  $\text{CsPbBr}_3$

Di Lei

Supervisor: Oleg. V. Prezhdo

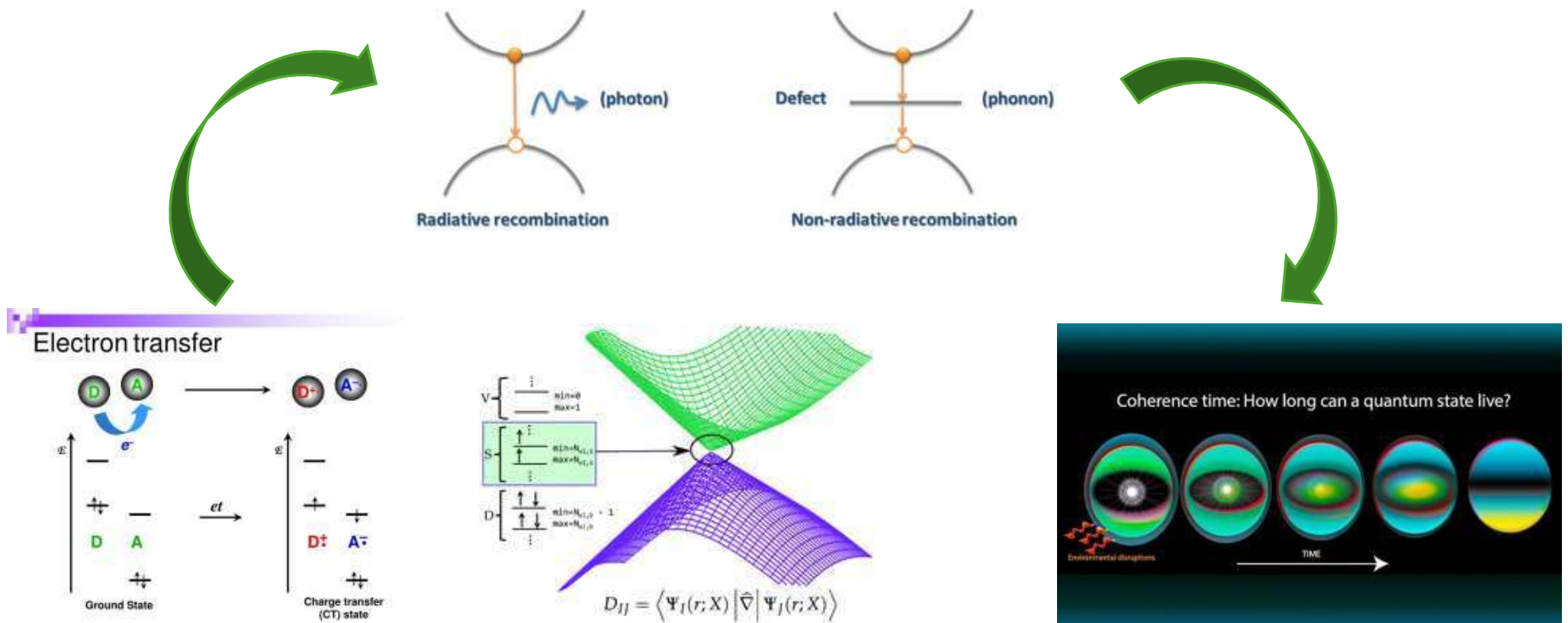
University of New Mexico

07/18/2026

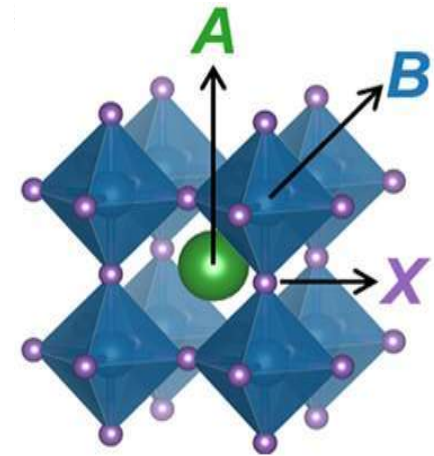
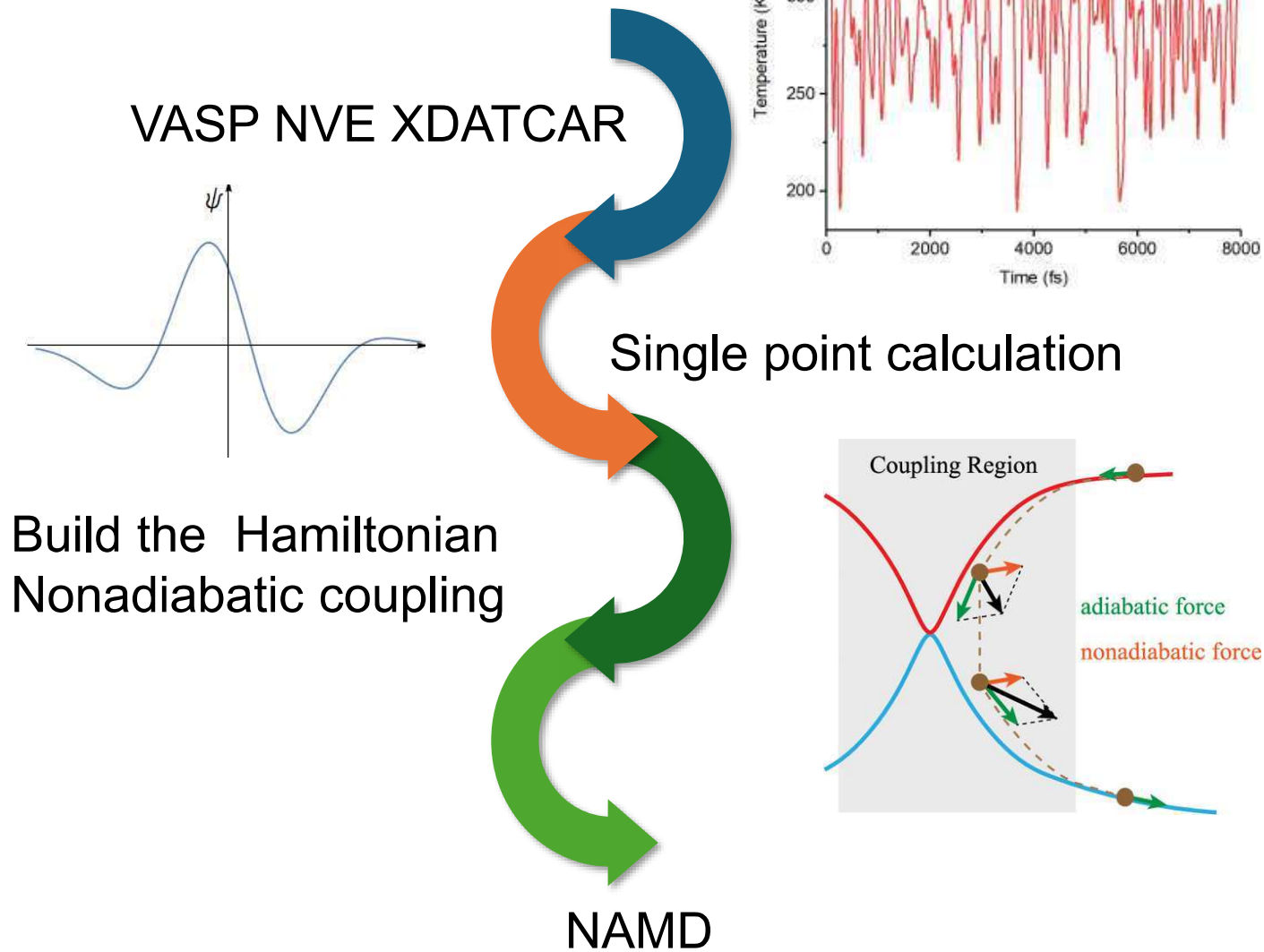


# Motivation

Nonadiabatic coupling describes the probability amplitude for electronic transitions induced by nuclear motion. (break down the Born-Oppenheimer approximation )

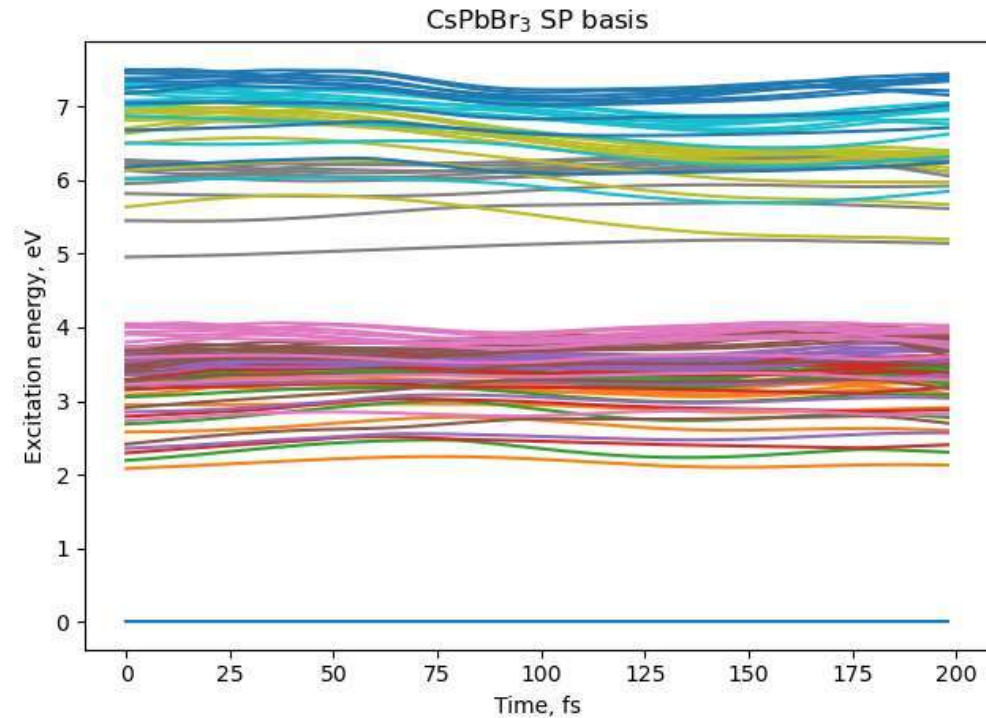


# Work Flow

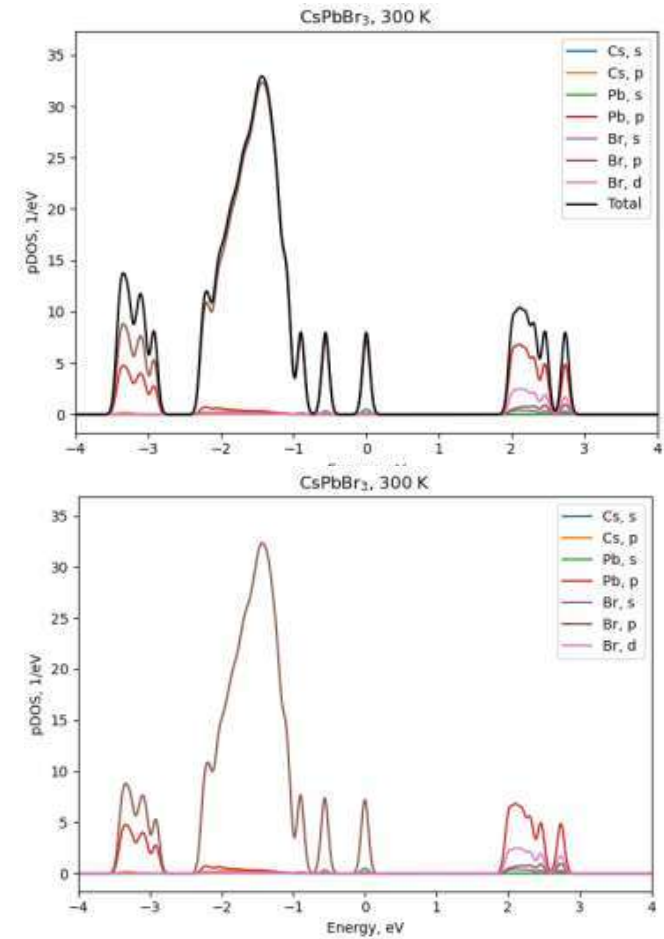


CsPbBr<sub>3</sub>

# Results and Discussion



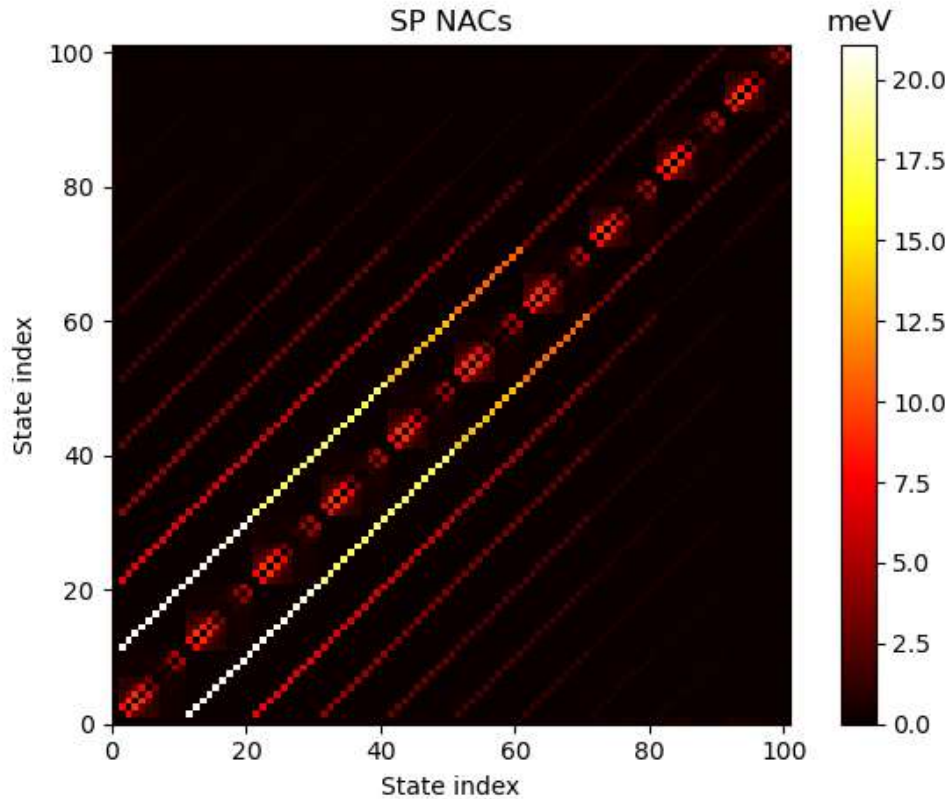
one clustering 2~4 eV, another one 5~7.5 eV,  
which is separated by a clear gap in between



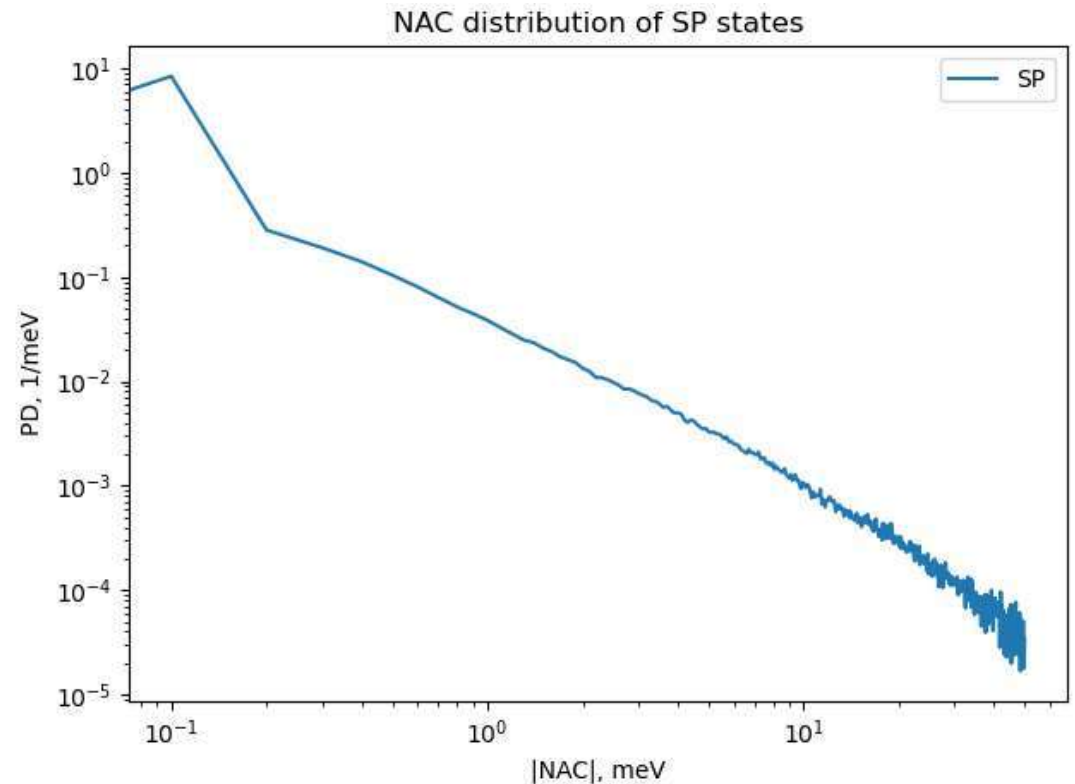
The band-edge states (VBM near -1.5 eV, CBM near 2 eV)  
are built almost entirely from Br p and Pb p orbitals.

# Results and Discussion

Where the coupling lives



Most couplings are weak, a few are not

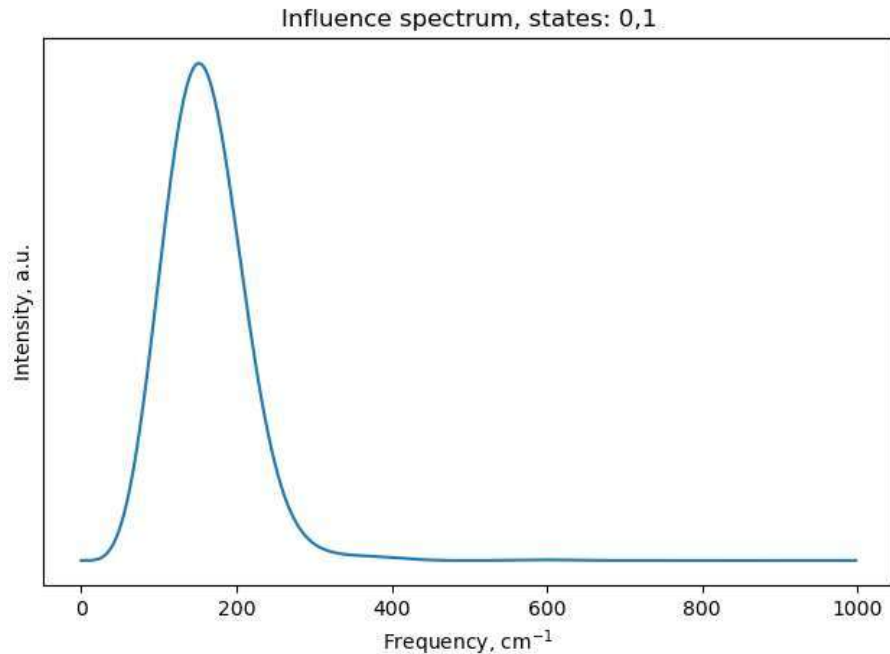


coupling concentrated tightly along the diagonal,  
adjacent energy states couple strongly (up to ~20 meV)  
nearby states talk to each other, distant ones barely do.



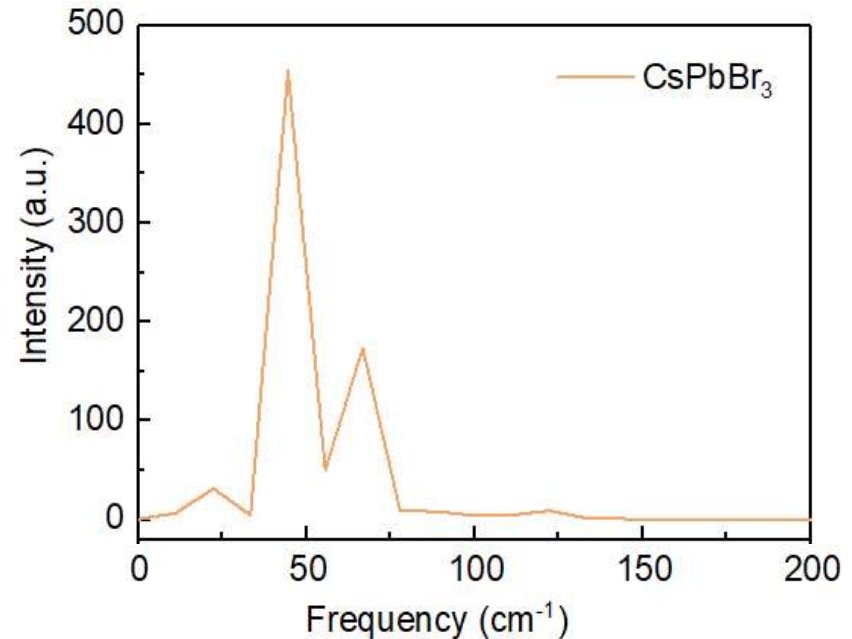
# Results and Discussion

phonons modes are participating the coupling



Pb-Br octahedral bending/tilting motions ( $\sim 200 \text{ cm}^{-1}$ )

it's the soft, floppy inorganic cage — not stiff internal vibrations — that modulates the electronic states and mediates coupling



The frequency at low temperature 3.2 K

Phonon modes:  $50 \text{ cm}^{-1}$ ,  $70 \text{ cm}^{-1}$

Lower temperature, the phonon modes tend to lower frequency

# Results and Discussion

## Coherence time

```
# ===== Computing the energy gaps and decoherence times
# Average decoherence times and rates
tau, rates = decoherence_times.decoherence_times_ave([Hvib_adi], [0], len(Hvib_adi), 0)

# Computes the energy gaps between all states for all steps
dE = decoherence_times.energy_gaps_ave([Ham_adi], [0], len(Ham_adi))

# Decoherence times in fs
deco_times = data_conv.MATRIX2nparray(tau) * units.au2fs

# Zero all the diagonal elements of the decoherence matrix
np.fill_diagonal(deco_times, 0)

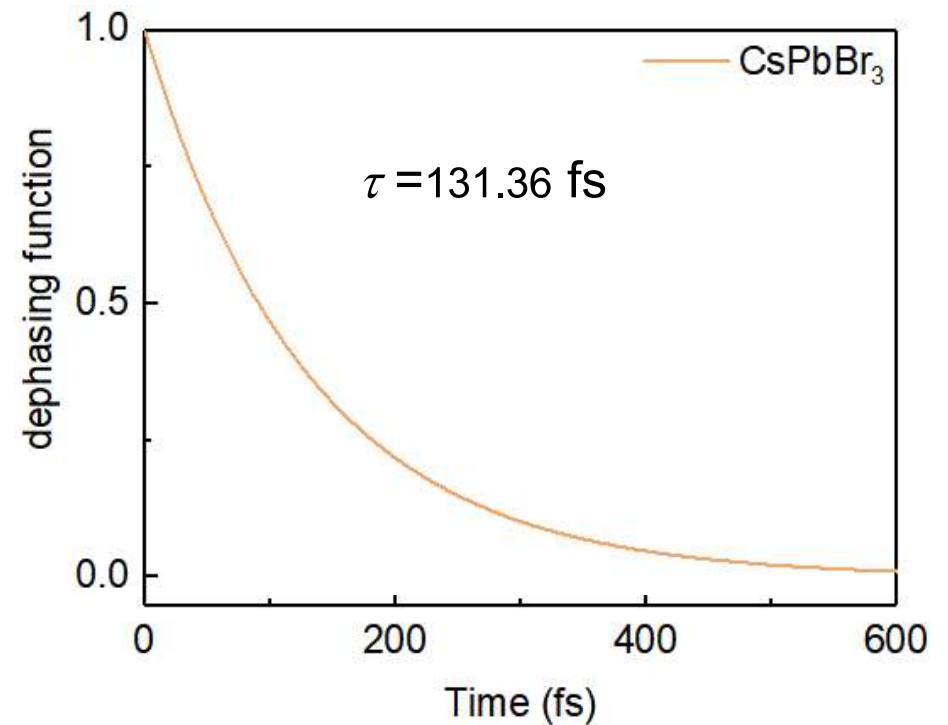
# Saving the average decoherence times
np.savetxt('decoherence_times.txt', deco_times.real)

# Computing the average energy gaps
gaps = MATRIX(NSTATES, NSTATES)
for step in range(NSTEPS):
    gaps += dE[step]
gaps /= NSTEPS

rates.show_matrix("decoherence_rates.txt")
gaps.show_matrix("average_gaps.txt")

deco = np.loadtxt('decoherence_times.txt')
print(F"GS <-> state {ISTATE}: {deco[0, ISTATE]:.1f} fs")

GS <-> state 91: 12.9 fs
```



The interesting part I have noticed is for the same material, the coherence time is 12.9 fs at 300K, while 131.36 fs at 3.2 K, which makes sense to me that low temperature can keep the quantum states longer

# Conclusion

CsPbBr<sub>3</sub>'s electronic structure is thermally robust: the SP excitation manifolds (2–4 eV and 5–7.5 eV) stay clearly separated over the full 300 K trajectory — nuclear motion perturbs but doesn't scramble the states

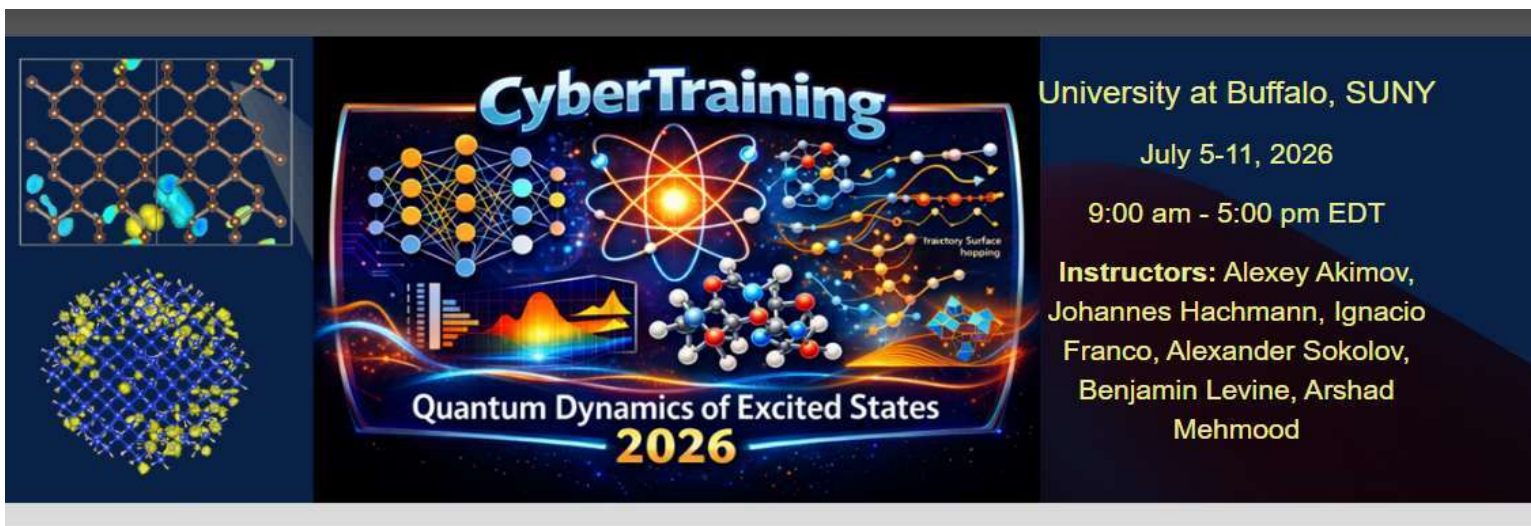
NACs follow an energy-gap law: strong coupling (up to ~20 meV) is confined to near-diagonal, near-degenerate state pairs, and the overall distribution is dominated by weak couplings with a long tail of rare strong ones — most state pairs are non-adiabatically "quiet."

The coupling is phonon-mediated through low-frequency (~200 cm<sup>-1</sup>) Pb-Br octahedral bending/tilting modes, not stiff internal vibrations or Cs motion — the soft, floppy inorganic cage is what drives electron-phonon coupling.

*Note: My NAMD script is still running, it will be done soon~*



# Thanks!



The banner is divided into three main sections. The left section features two molecular models: a 2D hexagonal lattice with blue and yellow spheres, and a 3D spherical cluster of blue and yellow spheres. The middle section is a vibrant collage with the text 'CyberTraining' in a stylized blue font at the top. Below it, there's a network diagram with orange and blue nodes, a glowing atomic model, a molecular structure, and a spectral plot. The text 'Quantum Dynamics of Excited States' and '2026' are prominently displayed at the bottom of this section. The right section has a dark blue background with white and yellow text providing event details.

University at Buffalo, SUNY  
July 5-11, 2026  
9:00 am - 5:00 pm EDT  
**Instructors:** Alexey Akimov,  
Johannes Hachmann, Ignacio  
Franco, Alexander Sokolov,  
Benjamin Levine, Arshad  
Mehmood